

# CSE260: Digital Logic Design

## Assignment 1

Summer 2025

Total marks: 5

### Question 1.

[2]

The engineering team on **Space Station Zeta** has reported two new types of malfunctions: **Malfunction Alpha** and **Malfunction Beta**. You are responsible for building a detection system to identify the presence of these malfunctions based on different sensor readings. Also, draw the circuit diagram.

A system may have no malfunction, any one of them, or both.

#### Inputs (Sensor Readings):

- **A: Oxygen Level Stable** (A=1 if oxygen level is stable, A=0 otherwise)
- **B: Internal Pressure Normal** (B=1 if pressure is normal, B=0 otherwise)
- **C: Power Supply Fluctuation** (C=1 if there's fluctuation, C=0 otherwise)
- **D: Communication Signal Loss** (D=1 if communication signal is lost, D=0 otherwise)

#### Outputs:

- **Alpha:** Malfunction Alpha
- **Beta:** Malfunction Beta

#### Logic Rules:

- You have **Malfunction Alpha** if the power supply fluctuation is observed along with:
  - (i) stable oxygen level **and** normal pressure, **or**
  - (ii) no signal loss.
- If you **don't** have power fluctuation, then Malfunction Alpha is triggered if:
  - (i) pressure is **not normal**, **or**
  - (ii) signal is lost **while oxygen is not stable**.
- You have **Malfunction Beta** if:
  - (i) the pressure is **not normal**, **or**
  - (ii) the signal is lost **while oxygen is not stable**.

**Question 2.****[1]**

Simplify the boolean equations using boolean algebra

a)  $AB + AB' + A'B$

b)  $(X + Y')(X + Y) + X'Y$

c)  $M'N + MN + M'N'$

d)  $(P + Q)' + (P' + Q)$

e)  $(RS)' + R + S'$

f)  $(A + B)(A' + C) + AB + AC$

**Question 3.****[1]**

Find the SOP for the function and express the function in full canonical SOP form:

$$F(A, B, C, D) = A'D + B'C$$

**Question 4.****[1]**

Find the POS for the function and express the function in full canonical POS form:

$$F(A, B, C, D) = A'D + B'C$$